human intuition, and its embodiment in classical physics, envision a reality in which things are always definitely one way *or* another, quantum mechanics describes a reality in which things sometimes hover in a haze of being partly one way *and* partly another. Things become definite only when a suitable observation forces them to relinquish quantum possibilities and settle on a specific outcome. The outcome that's realized, though, cannot be predicted—we can predict only the odds that things will turn out one way or another.

This, plainly speaking, is weird. We are unused to a reality that remains ambiguous until perceived. But the oddity of quantum mechanics does not stop here. At least as astounding is a feature that goes back to a paper Einstein wrote in 1935 with two younger colleagues, Nathan Rosen and Boris Podolsky, that was intended as an attack on quantum theory.[3] With the ensuing twists of scientific progress, Einstein's paper can now be viewed as among the first to point out that quantum mechanics— if taken at face value—implies that something you do over here can be *instantaneously* linked to something happening over there, regardless of distance. Einstein considered such instantaneous connections ludicrous and interpreted their emergence from the mathematics of quantum theory as evidence that the theory was in need of much development before it would attain an acceptable form. But by the 1980s, when both theoretical and technological developments brought experimental scrutiny to bear on these purported quantum absurdities, researchers confirmed that there *can* be an instantaneous bond between what happens at widely separated locations. Under pristine laboratory conditions, what Einstein thought absurd really happens (Chapter 4).

The implications of these features of quantum mechanics for our picture of reality are a subject of ongoing research. Many scientists, myself included, view them as part of a radical quantum updating of the meaning and properties of space. Normally, spatial separation implies physical independence. If you want to control what's happening on the other side of a football field, you have to go there, or, at the very least, you have to send someone or something (the assistant coach, bouncing air molecules conveying speech, a flash of light to get someone's attention, etc.) across the field to convey your influence. If you don't—if you remain spatially isolated—you will have no impact, since